Resultados Yolo:

Ultralytics 8.3.128 🚀 Python-3.11.12 torch-2.6.0+cu124 CUDA:0 (Tesla T4, 15095MiB)

[34m [1mengine/trainer: [0magnostic_nms=False, amp=True, augment=False, auto_augment=randaugment, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close_mosaic=10, cls=0.5, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=data/Multi-Class Animal Detection.v1-yolov8/animal-detection.yaml, degrees=0.0, deterministic=True, device=None, dfl=1.5, dnn=False, dropout=0.0, dynamic=False, embed=None, epochs=20, erasing=0.4, exist_ok=False, fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolo11s.pt, momentum=0.937, mosaic=1.0, multi_scale=False, name=animal_yolov11s, nbs=64, nms=False, opset=None, optimize=False, optimizer=auto, overlap_mask=True, patience=5, perspective=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=None, rect=False, resume=False, retina_masks=False, save=True, save_conf=False, save_crop=False, save_dir=runs/detect/animal_yolov11s, save_frames=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=0, shear=0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True, single_cls=False, source=None, split=val, stream_buffer=False, task=detect, time=None, tracker=botsort.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=0.0005, workers=8, workspace=None

Downloading https://ultralytics.com/assets/Arial.ttf to '/root/.config/Ultralytics/Arial.ttf'...

100%|[redacted]| 755k/755k [00:00<00:00, 21.8MB/s]Overriding model.yaml nc=80 with nc=20

	from	n	params	module	arguments
0	-1	1	928	ultralytics.nn.modules.conv.Conv	[3, 32, 3, 2]
1	-1	1	18560	ultralytics.nn.modules.conv.Conv	[32, 64, 3, 2]
2	-1	1	26080	ultralytics.nn.modules.block.C3k2	[64, 128, 1, False, 0.25]
3	-1	1	147712	ultralytics.nn.modules.conv.Conv	[128, 128, 3, 2]
4	-1	1	103360	ultralytics.nn.modules.block.C3k2	[128, 256, 1, False, 0.25]
5	-1	1	590336	ultralytics.nn.modules.conv.Conv	[256, 256, 3, 2]
6	-1	1	346112	ultralytics.nn.modules.block.C3k2	[256, 256, 1, True]
7	-1	1	1180672	ultralytics.nn.modules.conv.Conv	[256, 512, 3, 2]

```
  8                -1  1   1380352  ultralytics.nn.modules.block.C3k2         [512, 512, 1, True]
  9                -1  1    656896  ultralytics.nn.modules.block.SPPF         [512, 512, 5]
 10                -1  1    990976  ultralytics.nn.modules.block.C2PSA        [512, 512, 1]
 11                -1  1         0  torch.nn.modules.upsampling.Upsample       [None, 2, 'nearest']
 12           [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat         [1]
 13                -1  1    443776  ultralytics.nn.modules.block.C3k2         [768, 256, 1, False]
 14                -1  1         0  torch.nn.modules.upsampling.Upsample       [None, 2, 'nearest']
 15           [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat         [1]
 16                -1  1    127680  ultralytics.nn.modules.block.C3k2         [512, 128, 1, False]
 17                -1  1    147712  ultralytics.nn.modules.conv.Conv          [128, 128, 3, 2]
 18          [-1, 13]  1         0  ultralytics.nn.modules.conv.Concat         [1]
 19                -1  1    345472  ultralytics.nn.modules.block.C3k2         [384, 256, 1, False]
 20                -1  1    590336  ultralytics.nn.modules.conv.Conv          [256, 256, 3, 2]
 21          [-1, 10]  1         0  ultralytics.nn.modules.conv.Concat         [1]
 22                -1  1   1511424  ultralytics.nn.modules.block.C3k2         [768, 512, 1, True]
 23      [16, 19, 22]  1    827148  ultralytics.nn.modules.head.Detect         [20, [128, 256, 512]]
YOLO11s summary: 181 layers, 9,435,532 parameters, 9,435,516 gradients, 21.6 GFLOPs


Transferred 493/499 items from pretrained weights
Freezing layer 'model.23.dfl.conv.weight'
  [34m  [1mAMP:   [0mrunning Automatic Mixed Precision (AMP) checks...
Downloading https://github.com/ultralytics/assets/releases/download/v8.3.0/yolo11n.pt to 'yolo11n.pt'...
100%|██████████| 5.35M/5.35M [00:00<00:00, 101MB/s]
  [34m  [1mAMP:   [0mchecks passed ✅
  [34m  [1mtrain:   [0mFast image access ✅ (ping: 0.0±0.0 ms, read: 178.5±220.1 MB/s, size: 47.7 KB)
  [34m  [1mtrain:   [0mScanning /content/data/Multi-Class Animal Detection.v1-yolov8/train/labels... 1400 images, 1 backgrounds, 0 corrupt: 100%|██████████|
```

1400/1400 [00:01<00:00, 718.97it/s] [34m [1mtrain: [0mNew cache created: /content/data/Multi-Class Animal Detection.v1-yolov8/train/labels.cache

  [34m [1malbumentations: [0mBlur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average', num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))

  [34m [1mval: [0mFast image access ✅ (ping: 0.0±0.0 ms, read: 10.9±6.6 MB/s, size: 40.5 KB)

  [34m [1mval: [0mScanning /content/data/Multi-Class Animal Detection.v1-yolov8/valid/labels... 300 images, 1 backgrounds, 0 corrupt: 100%|[redacted]| 300/300 [00:00<00:00, 489.92it/s] [34m [1mval: [0mNew cache created: /content/data/Multi-Class Animal Detection.v1-yolov8/valid/labels.cache

Plotting labels to runs/detect/animal_yolov11s/labels.jpg...

  [34m [1moptimizer: [0m 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum' automatically...

  [34m [1moptimizer: [0m AdamW(lr=0.000417, momentum=0.9) with parameter groups 81 weight(decay=0.0), 88 weight(decay=0.0005), 87 bias(decay=0.0)

Image sizes 640 train, 640 val

Using 2 dataloader workers

Logging results to [1mruns/detect/animal_yolov11s [0m

Starting training for 20 epochs...

    Epoch   GPU_mem   box_loss   cls_loss   dfl_loss Instances   Size

     1/20   4.05G   1.035   3.799   1.511   34   640: 100%|[redacted]| 88/88 [00:32<00:00, 2.74it/s]

               Class   Images Instances   Box(P   R   mAP50 mAP50-95): 100%|[redacted]| 10/10 [00:05<00:00, 1.82it/s]   all   300   412   0.339   0.444   0.373   0.26

    Epoch   GPU_mem   box_loss   cls_loss   dfl_loss Instances   Size

2/20 4.85G 1.021 2.357 1.45 31 640: 100%|| 88/88 [00:27<00:00, 3.17it/s]

Class Images Instances Box(P R mAP50 mAP50-95): 100%|| 10/10 [00:02<00:00, 3.38it/s]

all 300 412 0.464 0.474 0.475 0.316

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

3/20 4.88G 1.038 2.076 1.479 25 640: 100%|| 88/88 [00:27<00:00, 3.17it/s]

Class Images Instances Box(P R mAP50 mAP50-95): 100%|| 10/10 [00:03<00:00, 3.00it/s] all 300 412 0.449 0.462 0.461 0.266

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

4/20 4.91G 1.056 1.976 1.482 33 640: 100%|| 88/88 [00:27<00:00, 3.19it/s]

Class Images Instances Box(P R mAP50 mAP50-95): 100%|| 10/10 [00:02<00:00, 3.39it/s] all 300 412 0.656 0.429 0.512 0.324

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

5/20 4.96G 1.061 1.802 1.47 18 640: 100%|| 88/88 [00:27<00:00, 3.18it/s]

Class Images Instances Box(P R mAP50 mAP50-95): 100%|| 10/10 [00:02<00:00, 3.35it/s] all 300 412 0.525 0.521 0.522 0.348

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

6/20 5G 1.016 1.678 1.455 28 640: 100%|| 88/88 [00:28<00:00, 3.13it/s]

```
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:03<00:00,  3.12it/s]                   all        300        412      0.567
0.608      0.636      0.406


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size

       7/20      5.04G      1.001      1.562      1.423         20        640: 100%|██████████| 88/88
[00:28<00:00,  3.10it/s]

                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  3.38it/s]                   all        300        412      0.592
0.561      0.651       0.44


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size

       8/20      5.08G     0.9613      1.494      1.417         30        640: 100%|██████████| 88/88
[00:27<00:00,  3.16it/s]

                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:03<00:00,  2.90it/s]                   all        300        412      0.721
0.521       0.62      0.418


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size

       9/20      5.12G     0.9321      1.437      1.379         51        640: 100%|██████████| 88/88
[00:27<00:00,  3.19it/s]

                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  3.39it/s]                   all        300        412      0.701
0.637      0.704      0.489


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size

      10/20      5.16G     0.9327      1.338      1.384         30        640: 100%|██████████| 88/88
[00:29<00:00,  3.02it/s]
```

Class Images Instances Box(P R mAP50 mAP50-95):
100%|██████████| 10/10 [00:03<00:00, 2.86it/s] all 300 412 0.717
0.612 0.682 0.489

Closing dataloader mosaic

[34m [1malbumentations: [0mBlur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average', num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

11/20 5.2G 0.8097 1.042 1.394 8 640: 100%|██████████| 88/88 [00:28<00:00, 3.10it/s]

Class Images Instances Box(P R mAP50 mAP50-95):
100%|██████████| 10/10 [00:03<00:00, 3.27it/s] all 300 412 0.745
0.641 0.727 0.51

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

12/20 5.24G 0.7745 0.8845 1.35 9 640: 100%|██████████| 88/88 [00:26<00:00, 3.29it/s]

Class Images Instances Box(P R mAP50 mAP50-95):
100%|██████████| 10/10 [00:03<00:00, 2.72it/s]

all 300 412 0.74 0.664 0.752 0.54

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

13/20 5.28G 0.7492 0.8645 1.341 9 640: 100%|██████████| 88/88 [00:26<00:00, 3.29it/s]

Class Images Instances Box(P R mAP50 mAP50-95):
100%|██████████| 10/10 [00:02<00:00, 3.35it/s] all 300 412 0.858
0.657 0.795 0.565

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

14/20 5.32G 0.7088 0.7322 1.296 9 640: 100%|| 88/88 [00:27<00:00, 3.19it/s]

Class Images Instances Box(P R mAP50 mAP50-95): 100%|| 10/10 [00:02<00:00, 3.41it/s] all 300 412 0.785 0.718 0.781 0.569

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

15/20 5.36G 0.6791 0.6534 1.265 8 640: 100%|| 88/88 [00:27<00:00, 3.24it/s]

Class Images Instances Box(P R mAP50 mAP50-95): 100%|| 10/10 [00:03<00:00, 3.15it/s] all 300 412 0.788 0.729 0.8 0.592

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

16/20 5.39G 0.6573 0.6213 1.245 8 640: 100%|| 88/88 [00:26<00:00, 3.28it/s]

Class Images Instances Box(P R mAP50 mAP50-95): 100%|| 10/10 [00:02<00:00, 3.39it/s] all 300 412 0.845 0.744 0.826 0.617

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

17/20 5.43G 0.6419 0.5661 1.231 9 640: 100%|| 88/88 [00:26<00:00, 3.26it/s]

Class Images Instances Box(P R mAP50 mAP50-95): 100%|| 10/10 [00:03<00:00, 2.89it/s] all 300 412 0.863 0.755 0.824 0.612

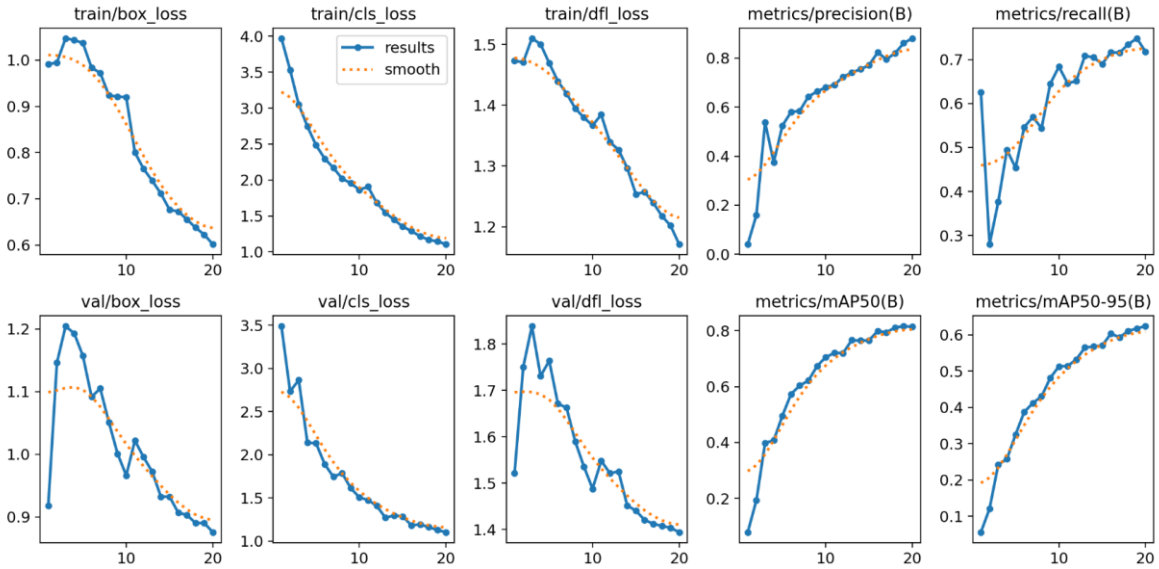
Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

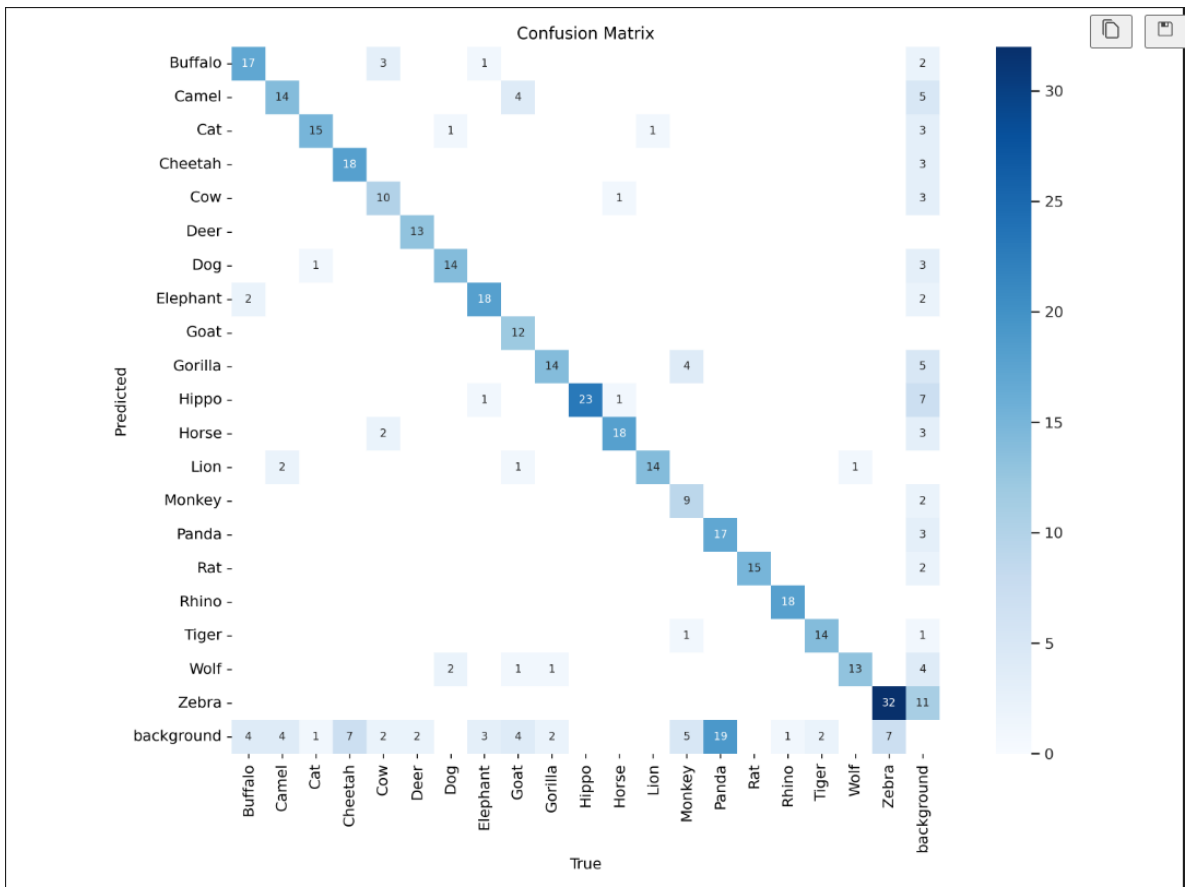
```
    18/20      5.47G    0.6193     0.509      1.21         9       640: 100%|██████████| 88/88
[00:27<00:00,  3.26it/s]
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  3.45it/s]                   all        300        412      0.823
0.76      0.849      0.637


    Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size

    19/20      5.52G    0.5846     0.4811      1.182        16       640: 100%|██████████| 88/88
[00:26<00:00,  3.27it/s]
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  3.43it/s]                   all        300        412      0.853
0.772      0.848      0.638


    Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size

    20/20      5.55G    0.5535     0.4468      1.138        13       640: 100%|██████████| 88/88
[00:27<00:00,  3.15it/s]
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:03<00:00,  2.82it/s]                   all        300        412      0.903
0.743      0.853      0.649


20 epochs completed in 0.176 hours.

Optimizer stripped from runs/detect/animal_yolov11s/weights/last.pt, 19.2MB

Optimizer stripped from runs/detect/animal_yolov11s/weights/best.pt, 19.2MB


Validating runs/detect/animal_yolov11s/weights/best.pt...

Ultralytics 8.3.128 🚀 Python-3.11.12 torch-2.6.0+cu124 CUDA:0 (Tesla T4, 15095MiB)

YOLO11s summary (fused): 100 layers, 9,420,540 parameters, 0 gradients, 21.3 GFLOPs

                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:04<00:00,  2.18it/s]
```

all	300	412	0.904	0.743	0.853	0.648
Buffalo	15	23	0.892	0.739	0.816	0.707
Camel	15	20	0.755	0.65	0.656	0.427
Cat	15	17	1	0.776	0.908	0.672
Cheetah	15	25	0.942	0.68	0.823	0.501
Cow	15	17	1	0.607	0.861	0.637
Deer	15	15	0.986	0.867	0.92	0.681
Dog	15	17	0.959	0.765	0.914	0.724
Elephant	15	23	0.946	0.756	0.895	0.717
Goat	15	22	0.912	0.471	0.757	0.592
Gorilla	15	17	0.745	0.882	0.8	0.633
Hippo	15	23	0.938	0.913	0.976	0.801
Horse	15	20	0.919	0.8	0.932	0.736
Lion	15	15	0.848	0.933	0.947	0.843
Monkey	15	19	0.934	0.316	0.681	0.434
Panda	15	36	0.9	0.389	0.566	0.404
Rat	15	15	0.857	0.867	0.939	0.686
Rhino	15	19	0.991	0.947	0.986	0.78
Tiger	15	16	1	0.874	0.971	0.643
Wolf	14	14	0.739	0.929	0.859	0.8
Zebra	15	39	0.821	0.706	0.85	0.552

Speed: 0.3ms preprocess, 4.8ms inference, 0.0ms loss, 3.3ms postprocess per image

Results saved to [1mruns/detect/animal_yolov11s [0m


train/box_loss
train/cls_loss
results
smooth
train/dfl_loss
metrics/precision(B)
metrics/recall(B)
val/box_loss
val/cls_loss
val/dfl_loss
metrics/mAP50(B)
metrics/mAP50-95(B)


Confusion Matrix
Predicted
True
Buffalo
Camel
Cat
Cheetah
Cow
Deer
Dog
Elephant
Goat
Gorilla
Hippo
Horse
Lion
Monkey
Panda
Rat
Rhino
Tiger
Wolf
Zebra
background

Resultados YOLO NANO:

Ultralytics 8.3.128 🚀 Python-3.11.12 torch-2.6.0+cu124 CUDA:0 (Tesla T4, 15095MiB)

[34m [1mengine/trainer: [0magnostic_nms=False, amp=True, augment=False, auto_augment=randaugment, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close_mosaic=10, cls=0.5, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=data/Multi-Class Animal Detection.v1-yolov8/animal-detection.yaml, degrees=0.0, deterministic=True, device=None, dfl=1.5, dnn=False, dropout=0.0, dynamic=False, embed=None, epochs=20, erasing=0.4, exist_ok=False, fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolo11n.pt, momentum=0.937, mosaic=1.0, multi_scale=False, name=animal_yolov11n, nbs=64, nms=False, opset=None, optimize=False, optimizer=auto, overlap_mask=True, patience=5, perspective=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=None, rect=False, resume=False, retina_masks=False, save=True, save_conf=False, save_crop=False, save_dir=runs/detect/animal_yolov11n, save_frames=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=0, shear=0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True, single_cls=False, source=None, split=val, stream_buffer=False, task=detect, time=None, tracker=botsort.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=0.0005, workers=8, workspace=None

Overriding model.yaml nc=80 with nc=20

	from	n	params	module	arguments
0	-1	1	464	ultralytics.nn.modules.conv.Conv	[3, 16, 3, 2]
1	-1	1	4672	ultralytics.nn.modules.conv.Conv	[16, 32, 3, 2]
2	-1	1	6640	ultralytics.nn.modules.block.C3k2	[32, 64, 1, False, 0.25]
3	-1	1	36992	ultralytics.nn.modules.conv.Conv	[64, 64, 3, 2]
4	-1	1	26080	ultralytics.nn.modules.block.C3k2	[64, 128, 1, False, 0.25]
5	-1	1	147712	ultralytics.nn.modules.conv.Conv	[128, 128, 3, 2]
6	-1	1	87040	ultralytics.nn.modules.block.C3k2	[128, 128, 1, True]
7	-1	1	295424	ultralytics.nn.modules.conv.Conv	[128, 256, 3, 2]
8	-1	1	346112	ultralytics.nn.modules.block.C3k2	[256, 256, 1, True]
9	-1	1	164608	ultralytics.nn.modules.block.SPPF	[256, 256, 5]

```
 10                -1  1    249728  ultralytics.nn.modules.block.C2PSA         [256, 256, 1]
 11                -1  1         0  torch.nn.modules.upsampling.Upsample       [None, 2, 'nearest']
 12           [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat         [1]
 13                -1  1    111296  ultralytics.nn.modules.block.C3k2          [384, 128, 1, False]
 14                -1  1         0  torch.nn.modules.upsampling.Upsample       [None, 2, 'nearest']
 15           [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat         [1]
 16                -1  1     32096  ultralytics.nn.modules.block.C3k2          [256, 64, 1, False]
 17                -1  1     36992  ultralytics.nn.modules.conv.Conv           [64, 64, 3, 2]
 18          [-1, 13]  1         0  ultralytics.nn.modules.conv.Concat         [1]
 19                -1  1     86720  ultralytics.nn.modules.block.C3k2          [192, 128, 1, False]
 20                -1  1    147712  ultralytics.nn.modules.conv.Conv           [128, 128, 3, 2]
 21          [-1, 10]  1         0  ultralytics.nn.modules.conv.Concat         [1]
 22                -1  1    378880  ultralytics.nn.modules.block.C3k2          [384, 256, 1, True]
 23      [16, 19, 22]  1    434572  ultralytics.nn.modules.head.Detect         [20, [64, 128, 256]]
YOLO11n summary: 181 layers, 2,593,740 parameters, 2,593,724 gradients, 6.5 GFLOPs


Transferred 448/499 items from pretrained weights
Freezing layer 'model.23.dfl.conv.weight'
  [34m  [1mAMP:   [0mrunning Automatic Mixed Precision (AMP) checks...
  [34m  [1mAMP:   [0mchecks passed ✅
  [34m  [1mtrain:   [0mFast image access ✅ (ping: 0.0±0.0 ms, read: 1559.1±393.0 MB/s, size: 47.7 KB)
  [34m  [1mtrain:   [0mScanning /content/data/Multi-Class Animal Detection.v1-yolov8/train/labels.cache... 1400 images, 1 backgrounds, 0 corrupt: 100%|━━━━━━━━━━━━| 1400/1400 [00:00<?, ?it/s]  [34m  [1malbumentations:   [0mBlur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average', num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))


  [34m  [1mval:   [0mFast image access ✅ (ping: 0.0±0.0 ms, read: 433.9±148.4 MB/s, size: 40.5 KB)
```

[34m [1mval: [0mScanning /content/data/Multi-Class Animal Detection.v1-yolov8/valid/labels.cache... 300 images, 1 backgrounds, 0 corrupt: 100%|██████████| 300/300 [00:00<?, ?it/s]

Plotting labels to runs/detect/animal_yolov11n/labels.jpg...

[34m [1moptimizer: [0m 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum' automatically...

[34m [1moptimizer: [0m AdamW(lr=0.000417, momentum=0.9) with parameter groups 81 weight(decay=0.0), 88 weight(decay=0.0005), 87 bias(decay=0.0)

Image sizes 640 train, 640 val

Using 2 dataloader workers

Logging results to [1mruns/detect/animal_yolov11n [0m

Starting training for 20 epochs...

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

1/20 2.71G 0.9911 3.971 1.473 34 640: 100%|██████████| 88/88 [00:27<00:00, 3.22it/s]

Class Images Instances Box(P R mAP50 mAP50-95): 100%|██████████| 10/10 [00:02<00:00, 3.50it/s] all 300 412 0.0403 0.625 0.0796 0.056

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

2/20 3.1G 0.9948 3.531 1.471 31 640: 100%|██████████| 88/88 [00:24<00:00, 3.62it/s]

Class Images Instances Box(P R mAP50 mAP50-95): 100%|██████████| 10/10 [00:03<00:00, 2.98it/s] all 300 412 0.16 0.28 0.194 0.121

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

3/20 3.1G 1.047 3.049 1.51 25 640: 100%|██████████| 88/88 [00:26<00:00, 3.33it/s]

```
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  4.06it/s]
                   all        300        412      0.538      0.376      0.398      0.242

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       4/20      3.1G      1.043      2.745       1.5         33       640: 100%|██████████| 88/88
[00:25<00:00,  3.52it/s]
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  4.10it/s]                   all        300        412      0.375
0.495      0.408      0.258


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       5/20      3.1G      1.037      2.483      1.469         18       640: 100%|██████████| 88/88
[00:25<00:00,  3.44it/s]
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  4.24it/s]                   all        300        412      0.525
0.454      0.496      0.325


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       6/20      3.1G     0.9834      2.296       1.44         28       640: 100%|██████████| 88/88
[00:24<00:00,  3.58it/s]
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  4.05it/s]
                   all        300        412       0.58      0.546      0.574      0.388


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       7/20      3.1G     0.9722      2.169      1.419         20       640: 100%|██████████| 88/88
[00:25<00:00,  3.49it/s]
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  3.86it/s]
                   all        300        412      0.585      0.569      0.604      0.412
```

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

8/20 3.1G 0.9237 2.02 1.395 30 640: 100%|└└└

```
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  4.14it/s]                   all        300        412      0.691
0.646      0.72      0.514


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size

      12/20      3.1G     0.7649      1.682      1.339          9        640: 100%|██████████| 88/88
[00:23<00:00,  3.69it/s]

                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  4.34it/s]                   all        300        412      0.723
0.651      0.72      0.532


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size

      13/20      3.1G      0.739      1.544      1.327          9        640: 100%|██████████| 88/88
[00:24<00:00,  3.60it/s]

                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:04<00:00,  2.00it/s]                   all        300        412      0.741
0.708      0.766      0.565


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size

      14/20      3.1G     0.7114      1.446      1.297          9        640: 100%|██████████| 88/88
[00:27<00:00,  3.19it/s]

                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  4.32it/s]                   all        300        412      0.755
0.705      0.764      0.568


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size

      15/20      3.1G     0.6764      1.352      1.254          8        640: 100%|██████████| 88/88
[00:23<00:00,  3.73it/s]
```

```
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  4.25it/s]                   all        300        412      0.772
0.69      0.766      0.571


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size

      16/20      3.1G     0.6727      1.288      1.257          8       640: 100%|██████████| 88/88
[00:24<00:00,  3.58it/s]

                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  4.21it/s]                   all        300        412      0.824
0.716      0.799      0.603


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size

      17/20      3.1G     0.6556      1.217      1.239          9       640: 100%|██████████| 88/88
[00:23<00:00,  3.71it/s]

                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  4.25it/s]                   all        300        412      0.795
0.715      0.794      0.594


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size

      18/20      3.1G     0.6377      1.167      1.218          9       640: 100%|██████████| 88/88
[00:23<00:00,  3.74it/s]

                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  4.23it/s]                   all        300        412       0.82
0.734      0.811      0.611


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size

      19/20      3.1G     0.6225      1.145      1.202         16       640: 100%|██████████| 88/88
[00:23<00:00,  3.75it/s]
```

```
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  4.34it/s]
                   all        300        412      0.862      0.748      0.816      0.618
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      20/20      3.1G     0.6016      1.105      1.171         13        640: 100%|██████████| 88/88 [00:23<00:00,  3.72it/s]
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:02<00:00,  4.26it/s]
                   all        300        412      0.881      0.718      0.814      0.624
```

20 epochs completed in 0.156 hours.

Optimizer stripped from runs/detect/animal_yolov11n/weights/last.pt, 5.5MB

Optimizer stripped from runs/detect/animal_yolov11n/weights/best.pt, 5.5MB

Validating runs/detect/animal_yolov11n/weights/best.pt...

Ultralytics 8.3.128 🚀 Python-3.11.12 torch-2.6.0+cu124 CUDA:0 (Tesla T4, 15095MiB)

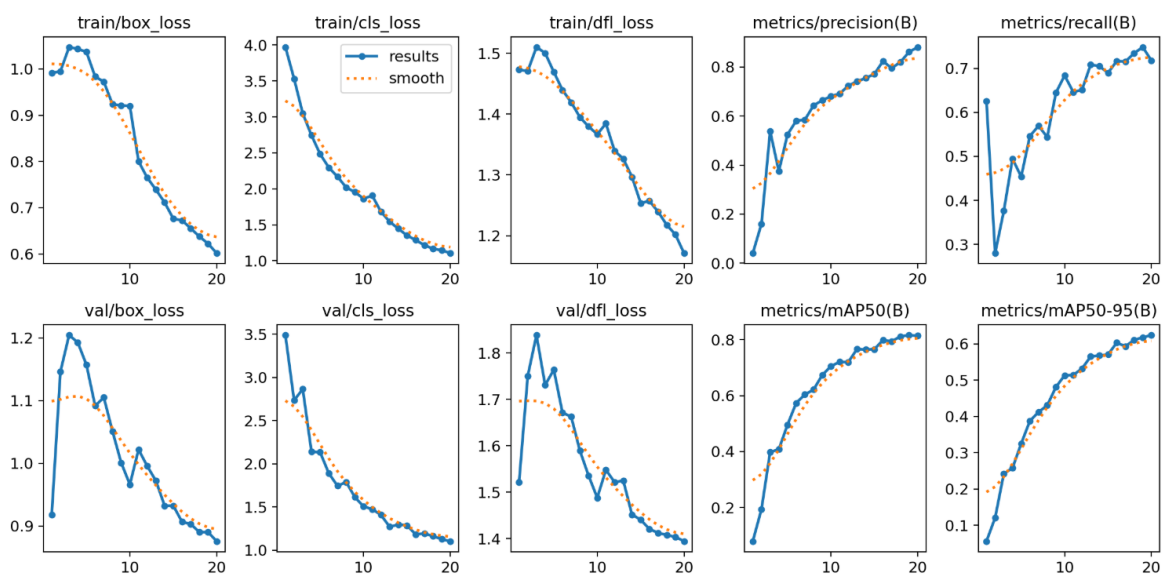
YOLO11n summary (fused): 100 layers, 2,586,052 parameters, 0 gradients, 6.3 GFLOPs

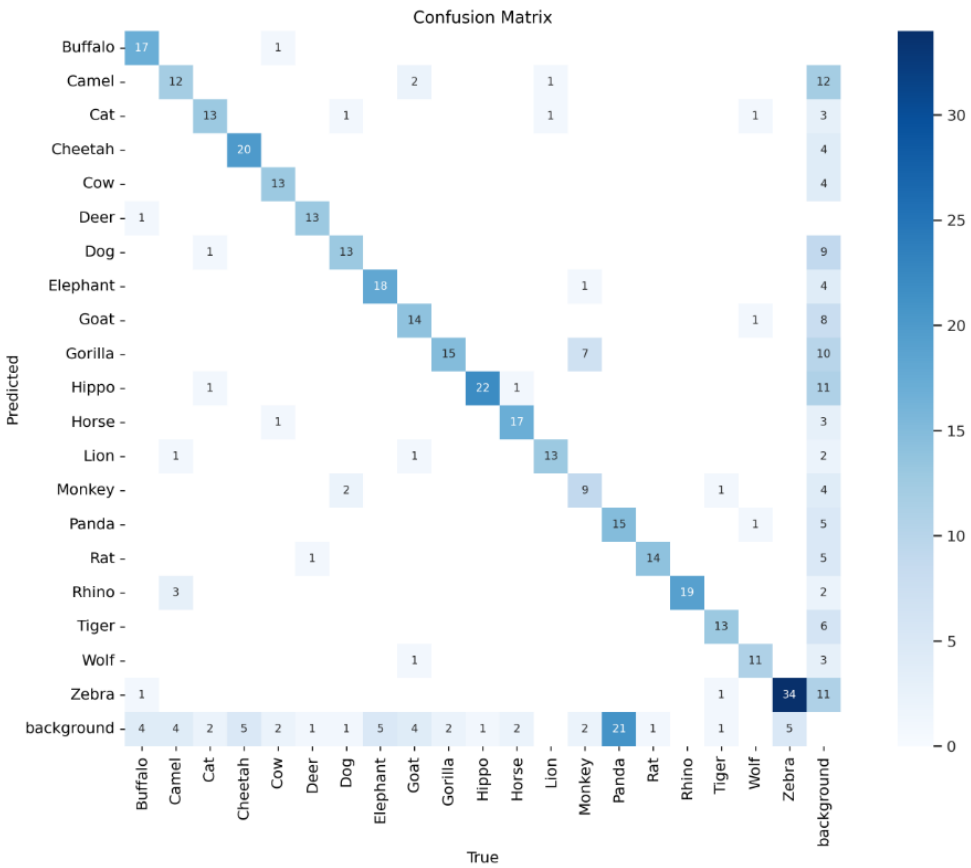
```
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100%|██████████| 10/10 [00:05<00:00,  1.91it/s]
                   all        300        412      0.878      0.719      0.814      0.624
              Buffalo         15         23      0.935      0.739       0.78       0.64
                Camel         15         20      0.709       0.55      0.637      0.482
                  Cat         15         17      0.721      0.765      0.804       0.58
              Cheetah         15         25          1      0.627      0.843      0.516
                  Cow         15         17      0.931      0.824      0.863      0.682
                 Deer         15         15      0.923      0.733       0.89      0.721
                  Dog         15         17          1       0.93      0.974      0.753
             Elephant         15         23      0.952      0.696      0.821      0.667
```

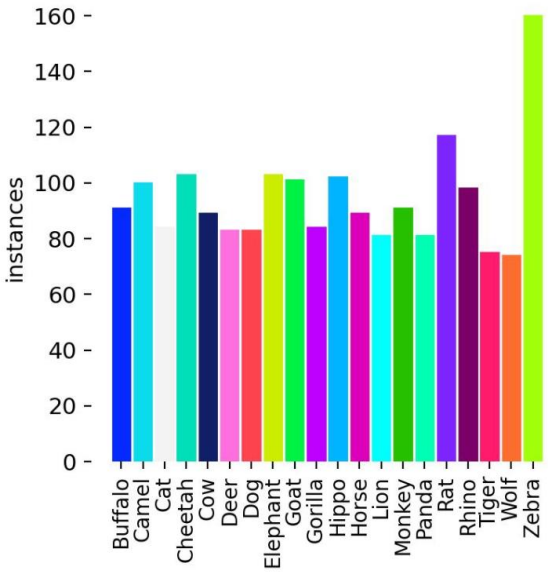
Goat	15	22	0.839	0.474	0.748	0.529
Gorilla	15	17	0.52	0.706	0.647	0.439
Hippo	15	23	0.844	0.826	0.892	0.645
Horse	15	20	0.945	0.852	0.951	0.777
Lion	15	15	0.923	0.867	0.904	0.794
Monkey	15	19	0.879	0.263	0.516	0.324
Panda	15	36	0.929	0.417	0.455	0.324
Rat	15	15	0.761	0.867	0.88	0.722
Rhino	15	19	0.947	0.934	0.975	0.813
Tiger	15	16	1	0.869	0.948	0.745
Wolf	14	14	0.901	0.653	0.875	0.766
Zebra	15	39	0.891	0.795	0.87	0.553

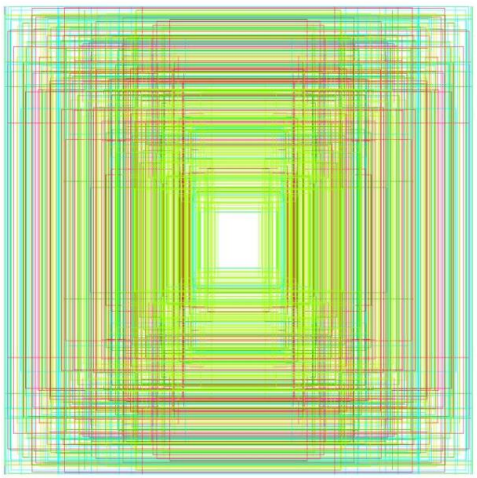
Speed: 0.3ms preprocess, 3.0ms inference, 0.0ms loss, 4.0ms postprocess per image

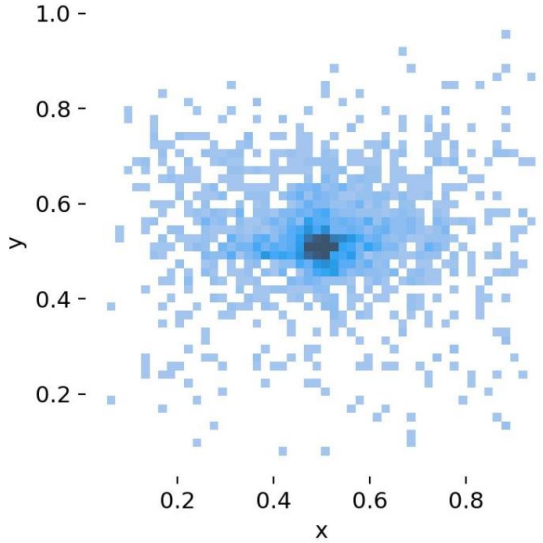
Results saved to [1mruns/detect/animal_yolov11n [0m

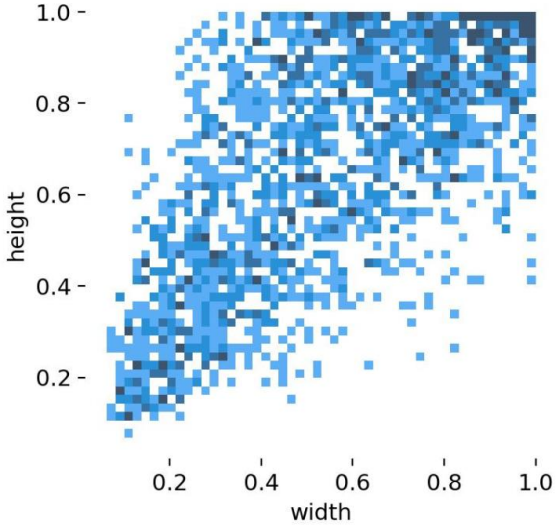



Confusion Matrix
Predicted
True
Buffalo
Camel
Cat
Cheetah
Cow
Deer
Dog
Elephant
Goat
Gorilla
Hippo
Horse
Lion
Monkey
Panda
Rat
Rhino
Tiger
Wolf
Zebra
background
0
5
10
15
20
25
30


instances
0
20
40
60
80
100
120
140
160
Buffalo
Camel
Cat
Cheetah
Cow
Deer
Dog
Elephant
Goat
Gorilla
Hippo
Horse
Lion
Monkey
Panda
Rat
Rhino
Tiger
Wolf
Zebra




y
0.2
0.4
0.6
0.8
1.0
x
0.2
0.4
0.6
0.8


height
0.2
0.4
0.6
0.8
1.0
width
0.2
0.4
0.6
0.8
1.0